EXP 3 I2C-BASED HEALTH MONITORING NODE FOR PULSE AND TEMPERATURE ACQUISITION

CODE:

```
#include <Wire.h>

const int pulsePin = 2;
volatile unsigned long pulseCount = 0;

void pulseDetected() {
  pulseCount++;
}

void setup() {
  Serial.begin(9600);
  Wire.begin();

  pinMode(pulsePin, INPUT);
  attachInterrupt(digitalPinToInterrupt(pulsePin),
                  pulseDetected, RISING);

  Serial.println("I2C HEALTH MONITORING");
}

void loop() {
  Wire.beginTransmission(0x48);
  Wire.write(0x00);
  byte error = Wire.endTransmission();
```

```
if (error == 0) {
  Wire.requestFrom(0x48, 2);

  if (Wire.available() >= 2) {
    int msb = Wire.read();
    int lsb = Wire.read();

    int raw = (msb << 8) | lsb;
    raw >>= 5;

    if (raw & 0x400) {
      raw -= 2048;
    }

    float temperature = raw * 0.125;

    noInterrupts();
    unsigned long count = pulseCount;
    pulseCount = 0;
    interrupts();

    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.println(" C");

    Serial.print("Pulse count in 10 sec: ");
    Serial.println(count);

    Serial.print("Pulse rate: ");
```

```
      Serial.print(count * 6);
      Serial.println(" BPM");
    }
  } else {
    Serial.println("LM75AD not detected");
  }

  delay(10000);
}
```

OUTPUT :

